Status	Finished
Started	Wednesday, 21 October 2026, 12:10 PM
Completed	Wednesday, 21 October 2026, 12:14 PM
Duration	3 mins 56 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Dividing the list in half repeatedly
b. Sorting the list before searching
c. Checking each element sequentially ✓
d. Skipping every second element

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search divides the list into halves.
b. Linear search can be applied to both sorted and unsorted lists. ✓
c. Linear search requires the list to be sorted.
d. Linear search is more efficient than binary search on large lists.

Question **3**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the search algorithm finishes
b. When the list contains duplicate elements
c. When the list is sorted
d. When the element is found in the list ✓

Question **4**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

- a. When the list is small and unsorted ✓
- b. When the list is small and sorted
- c. When the list is large and sorted
- d. When the list is large and unsorted

Question **5**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

- a. 89 and 94
- b. 90 and 94
- c. 90 and 99 ✓
- d. 89 and 99

Question **6**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

- a. By dividing the list length by two ✓
- b. By starting from the first element
- c. By comparing each element sequentially
- d. By using a hash function

Question **7**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

- a. O👎 ✓
- b. O(n log n)
- c. O(log n)
- d. O(1)

Question **8**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n-1
b. n ✓
c. log n
d. n/2

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. There must be a mechanism to access middle element directly
b. Binary search algorithm is not efficient when the data elements more than 1500 ✓
c. Must use a sorted array
d. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

Question **10**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Insertion

b. Complexity

c. Selection

d. Bubble ✓

Question **11**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By dividing the list into halves
b. By comparing each element in the list sequentially ✓
c. By using a hash function
d. By sorting the list first

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is very small
b. When the list is unsorted
c. When the list contains duplicate elements
d. When the list is sorted ✓

Question **13**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Hash search
b. Binary search ✓
c. Linear search
d. Sequential search

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. There should be direct access to the middle element in any sublist
b. Number values should only be present
Number values should only be present
c. The list must be sorted
d. There must be a mechanism to delete and/or insert elements in the list ✓

Question **15**

Incorrect

Mark 0.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

- a. Hash search ✕
- b. Binary search
- c. Linear search
- d. Interpolation search